

ABO Genotype, ‘Blood-Type’ Diet and Cardiometabolic Risk Factors



Background The ‘Blood-Type’ diet advises individuals to eat according to their ABO blood group to improve their health and decrease risk of chronic diseases such as cardiovascular disease. However, the association between blood type-based dietary patterns and health outcomes has not been examined. The objective of this study was to determine the association between ‘blood-type’ diets and biomarkers of cardiometabolic health and whether an individual's ABO genotype modifies any associations. **Methods** Subjects (n = 1,455) were participants of the Toronto Nutrigenomics and Health study. Dietary intake was assessed using a one-month, 196-item food frequency questionnaire and a diet score was calculated to determine relative adherence to each of the four ‘Blood-Type’ diets. ABO blood group was determined by genotyping rs8176719 and rs8176746 in the ABO gene. ANCOVA, with age, sex, ethnicity, and energy intake as covariates, was used to compare cardiometabolic biomarkers across tertiles of each ‘Blood-Type’ diet score. **Results** Adherence to the Type-A diet was associated with lower BMI, waist circumference, blood pressure, serum cholesterol, triglycerides, insulin, HOMA-IR and HOMA-Beta ($P < 0.05$). Adherence to the Type-AB diet was also associated with lower levels of these biomarkers ($P < 0.05$), except for BMI and waist circumference. Adherence to the Type-O diet was associated with lower triglycerides ($P < 0.0001$). Matching the ‘Blood-Type’ diets with the corresponding blood group did not change the effect size of any of these associations. No significant association was found for the Type-B diet. **Conclusions** Adherence to certain ‘Blood-Type’ diets is associated with favorable effects on some cardiometabolic risk factors, but these associations were independent of an individual's ABO genotype, so the findings do not support the ‘Blood-Type’ diet hypothesis.